Program 7: Write a program to generate in the Fibonacci series and store it in a list. Then find the sum of all values.

```
a=-1
b=1
n=int(input("Enter no. of terms: "))
i=0
sum=0
Fibo=[]
while i<n:
        s = a + b
        Fibo.append(s)
        sum+=s
        a = b
        b = s
        i+=1
print("Fibonacci series upto "+ str(n) +" terms is : " + str(Fibo))
print("The sum of Fibonacci series: ",sum)
```

Output

```
Enter no. of terms: 10
Fibonacci series upto 10 terms is : [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
The sum of Fibonacci series:  88
```

9.2 Tuples

Introduction to Tuples

Tuples consists of a number of values separated by comma and enclosed within parentheses. Tuple is similar to list, values in a list can be changed but not in a tuple.

The term Tuple is originated from the Latin word represents an abstraction of the sequence of numbers:

single(1), double(2), triple(3), quadruple(4), quintuple(5), sextuple(6), septuple(7), octuple(8), ..., n-tuple, ...,

9.2.1 Comparison of Tuples and list

1. The elements of a list are changeable (mutable) whereas the elements of a tuple are unchangeable (immutable), this is the key difference between tuples and list.
2. The elements of a list are enclosed within square brackets. But, the elements of a tuple are enclosed by paranthesis.
3. Iterating tuples is faster than list.

12th Computer Science_EM Chapter 9.indd 150 21-12-2022 15:37:02